

Draft Introduction Assignment

Problem Definition

Manga pages contain highly stylized and irregular visual structures, including non-rectangular panels, speech bubbles, sound effects, and artwork that frequently extends beyond panel boundaries. Unlike word-processing documents or conventional web pages, manga does not follow a predictable grid or consistent reading structure. Characters, speech bubbles, and decorative elements may overlap multiple panels or interrupt panel borders entirely. The research problem we are defining is how to automatically identify and associate manga panels primarily and speech bubbles even when the page structure cannot be inferred from complete borders or regular layouts due to art stylistic features. This is a difficult computer vision problem because existing document-layout systems assume that regions are rectangular, separated, and fully enclosed.

This project contributes to three broader areas of computer vision. First, it contributes to document and layout analysis, which studies how to segment and interpret complex pages. Second, it contributes to segmentation under partial occlusion, where objects must still be recognized even when their boundaries are hidden or incomplete. In manga, panel borders are often interrupted by speech bubbles or artwork, but the disconnected regions must still be recognized as belonging to the same panel. Third, it contributes to semantic image segmentation, specifically the task of dividing an image into meaningful regions when those regions do not have clear outlines.

Motivation and Value

This problem is important because many downstream applications depend on correctly identifying the individual components of a manga page. Applications require accurate locations of panels, speech bubbles, and text regions. One example is automatic translation, where the system must know which speech bubbles belong to which panel before translating the text in the correct order. Another example is reconstructing a digital representation of a manga page in which each panel, speech bubble, and text region exists

as a separate editable layer. Such a representation would support applications including accessible digital readers, scanlation assistance tools, and interfaces that allow users to manipulate or reformat manga content. Improving the ability to parse highly stylized documents would benefit accessibility technologies by making visual media easier to search, translate, and present in alternative formats.

Limitations of Existing Layout-Analysis Methods

The technical gap in existing work is that current methods rely either on spatial proximity or on simple border structure. Neither assumption is sufficient for highly stylized manga pages. Bajammal and Mesbah's work on webpage segmentation method groups nearby visual elements into the same region based on their spatial proximity. This assumption is effective for web pages because they typically follow an organized layout. However, it does not transfer well to manga. Two adjacent regions may belong to entirely different panels. As a result, approaches based only on proximity may incorrectly merge unrelated regions or split a single panel into multiple pieces.

Ho et al. improves upon earlier connected component and border detection techniques by repairing incomplete panel borders and recursively splitting the page. However, their approach still assumes that panels can be separated primarily through horizontal and vertical gaps in the page. This assumption works for relatively conventional manga layouts, but not when panels are circular, heavily overlapping, or interrupted by large visual elements. In these cases, the method may incorrectly split a single panel into multiple regions or merge multiple panels together.

High Level Methods/Approach

The system we propose will use a multi-stage approach to a manga parser by separating the identification of geometric and semantic cues.

The input will be a grayscale image of a manga page. Many manga panels use background smudging, grading, or shading effects. These effects are ornamental and don't help to differentiate between panels, and we don't want them to interfere with

identifying lines or semantic cues. So, we will first preprocess the input using binarization and noise reduction to emphasize the line art and suppress these background structures.

The first stage will be to identify panel boundaries using edge-based methods. We will apply a Canny edge detector to extract strong edges corresponding to panel borders. Edges from Canny are often fragmented with gaps and noise, so we will use morphological operations (dilation, closing) to fix broken lines. After this, panel borders are more likely to form continuous loops instead of broken segments. These contours are then filtered based on geometric properties (area, aspect ratio, and closure) to identify candidate panel regions. Manga panels are rarely exactly rectangular, and can at times also be more than four-sided, so polygonal approximations will be used to preserve irregular shapes.

In the second stage, the system performs semantic element detection within the detected panel regions. A pretrained object detection model (YOLO) is fine-tuned on annotated manga datasets (Manga109, MangaSeg) to identify speech bubbles. Detected objects will be assigned to panels based on spatial overlap, which captures which panel they visually belong to. However, because speech bubbles may reference characters across panel boundaries, this assignment can be further augmented by identifying the tail direction of the speech bubble to better model semantic relationships.

Finally, the outputs of these structural and semantic stages are combined into a hierarchical representation of the page. Each panel will be represented as a region containing associated objects, and adjacency relationships between panels will be computed to support downstream tasks such as reading order inference. This design allows each component to be improved independently while maintaining a coherent overall representation.

Expectations

We expect the system to accurately identify manga panels and adjacency relationships across a range of manga styles. These will include irregular panel layouts, overlapping panels, speech bubbles and characters across distinct panels, panels with heavy background artifacts, and non-linear arrangements. In addition to panel segmentation, the system should achieve reliable detection of speech bubbles within panels, enabling a more complete understanding of page structure.

Compared to approaches based purely on adjacency or clustering, we expect our model to perform better with irregular layouts where visual grouping does not strictly correspond to semantic grouping. For example, the system should correctly separate adjacent panels that share borders while still associating spatially separated elements within the same panel.

Works Cited

- [1]Ho, Ka-Chun, Cheuk-Man Leung, and Pong C. Yuen. “A Robust Panel Extraction
[2]Method for Manga.” Proceedings of the ACM International Conference on
Multimedia, 2014, pp. 1125–1128.

Bajammal, Mohammad, and Ali Mesbah. “Page Segmentation Using Visual Adjacency
Analysis.” *arXiv*, 23 Dec. 2021, arXiv:2112.11975.

GenAI Usage:

Used ChatGPT to think of a second downstream application that relies on accuracy of
panel parsing.